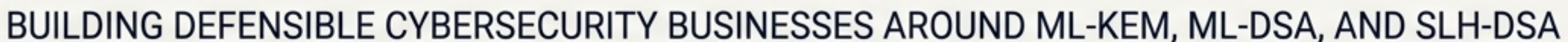
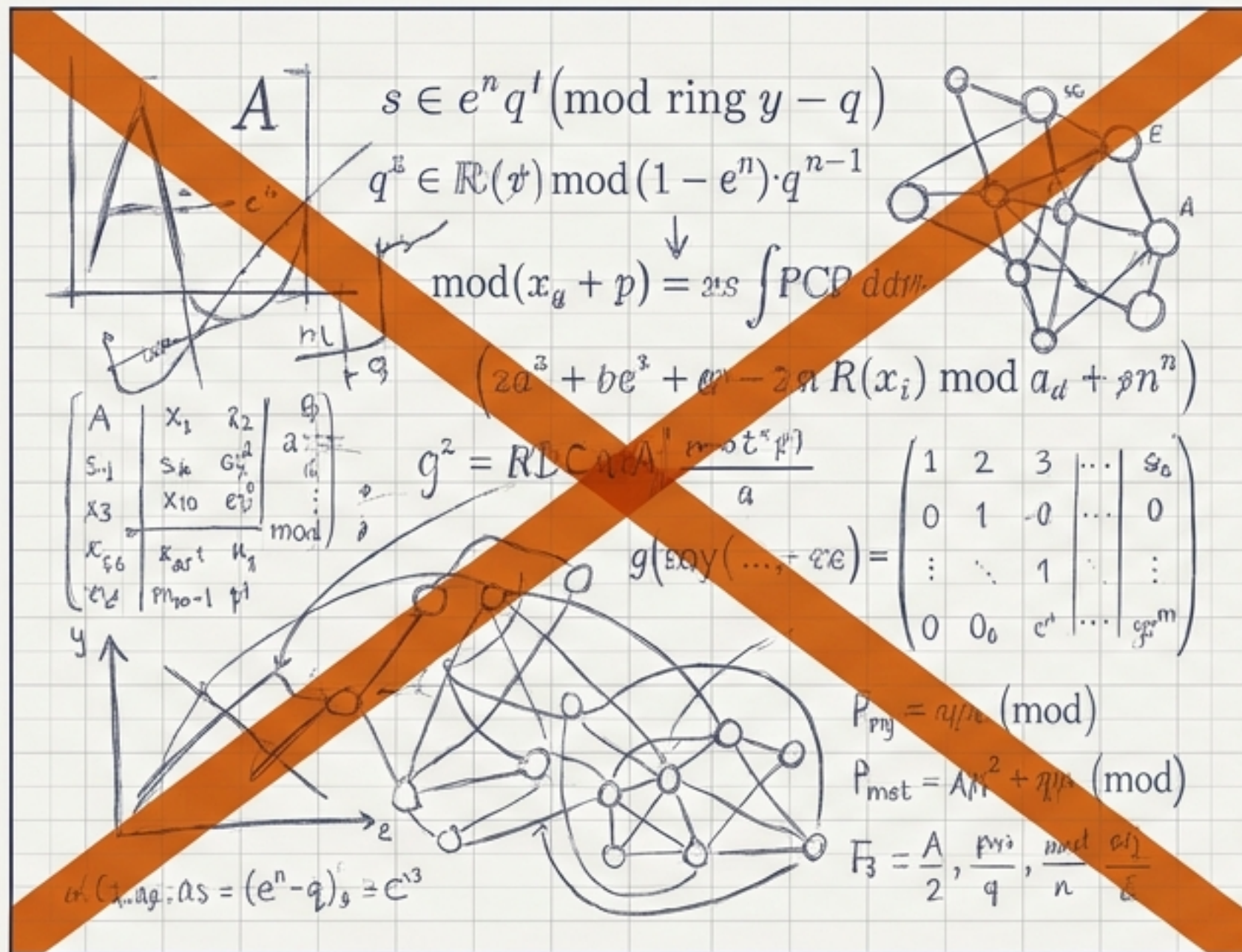


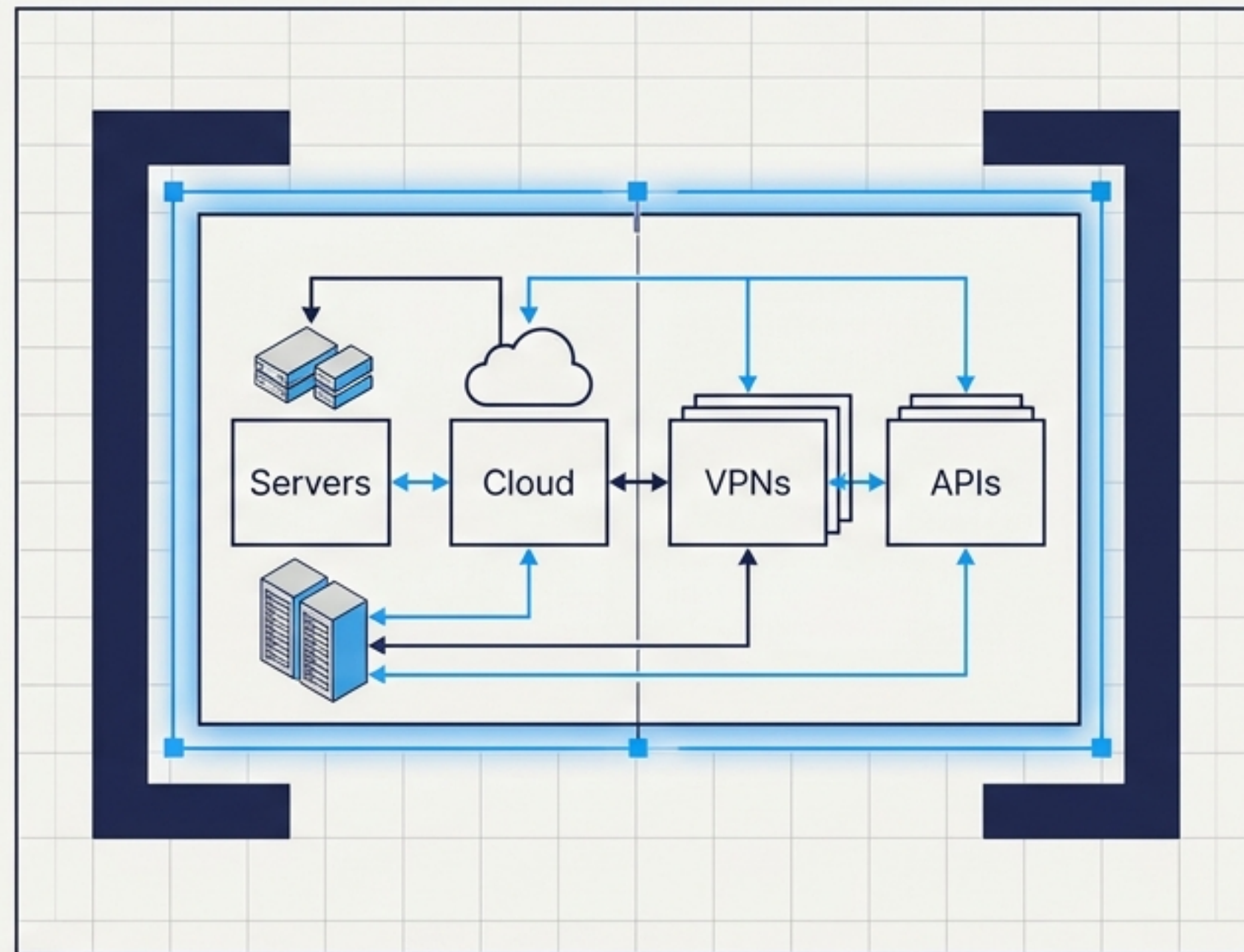
A Strategic Playbook and Market Map for the PQC Transition



Enterprises Do Not Buy Math. They Buy Security Outcomes.



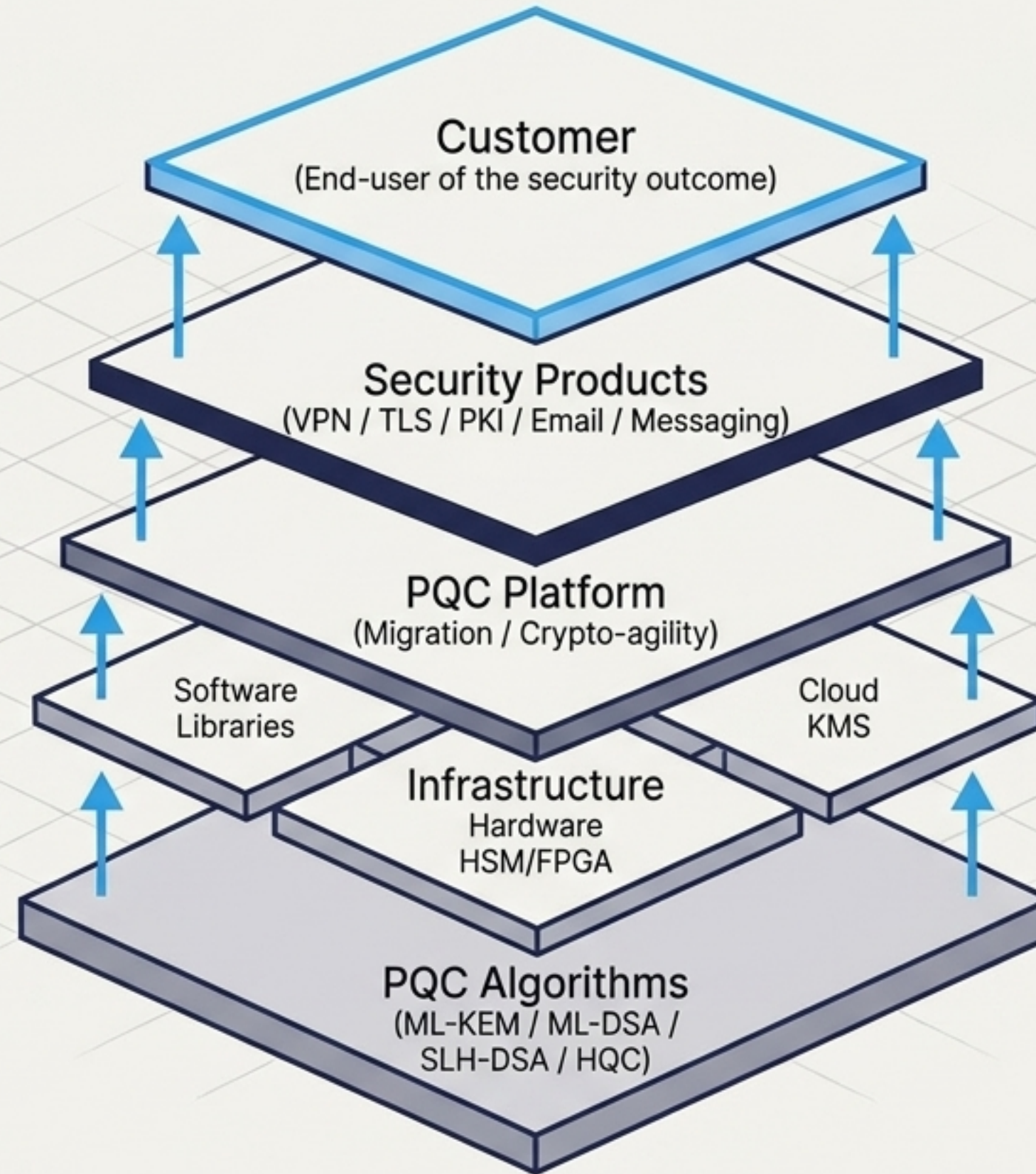
ML-KEM as a mathematical object is a public standard.
A plain implementation is nearly impossible to differentiate.



Organizations buy implementation, hardware, integration, migration, performance, and compliance.

The NIST Mandate: Identify vulnerable cryptography, plan replacement, update products, and build crypto-agility.

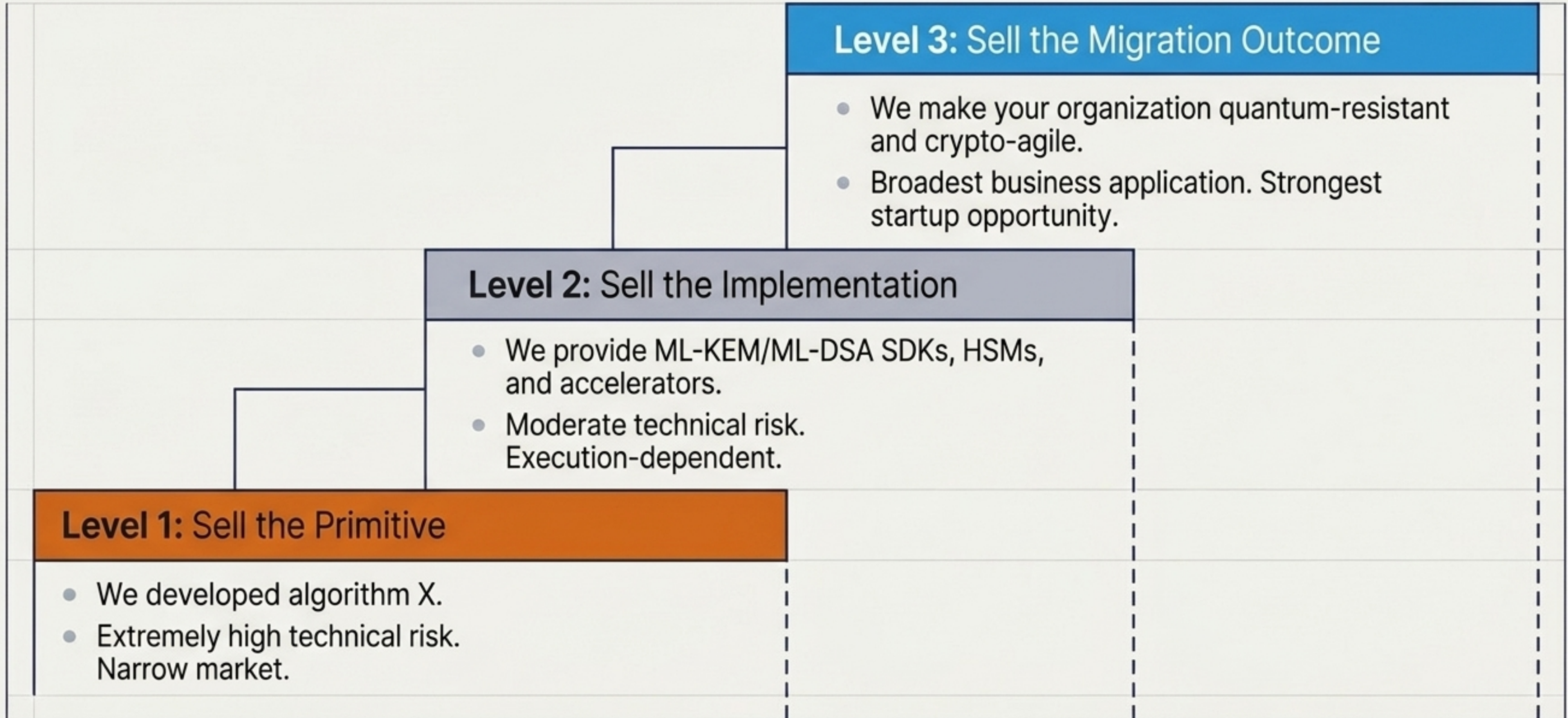
The PQC Commercialization Stack



A startup can operate at any single layer, or vertically integrate across several.

The Three Levels of Selling PQC

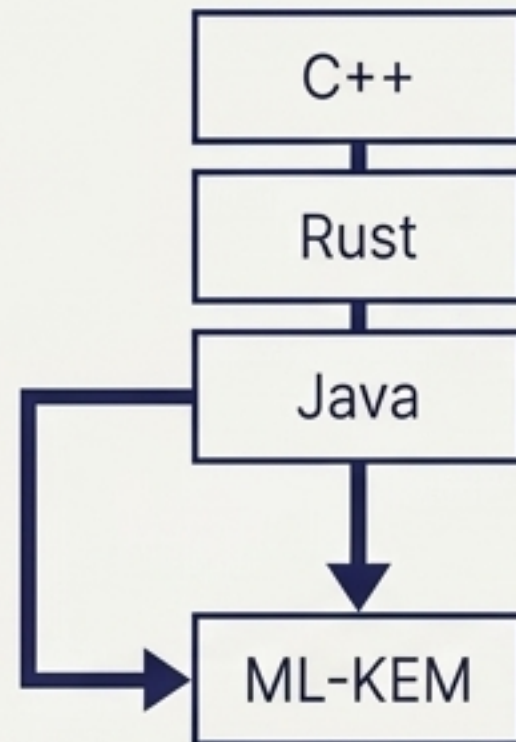
High Value



Pathway 1: The Implementers (Libraries & SDKs)

Direct Library Model

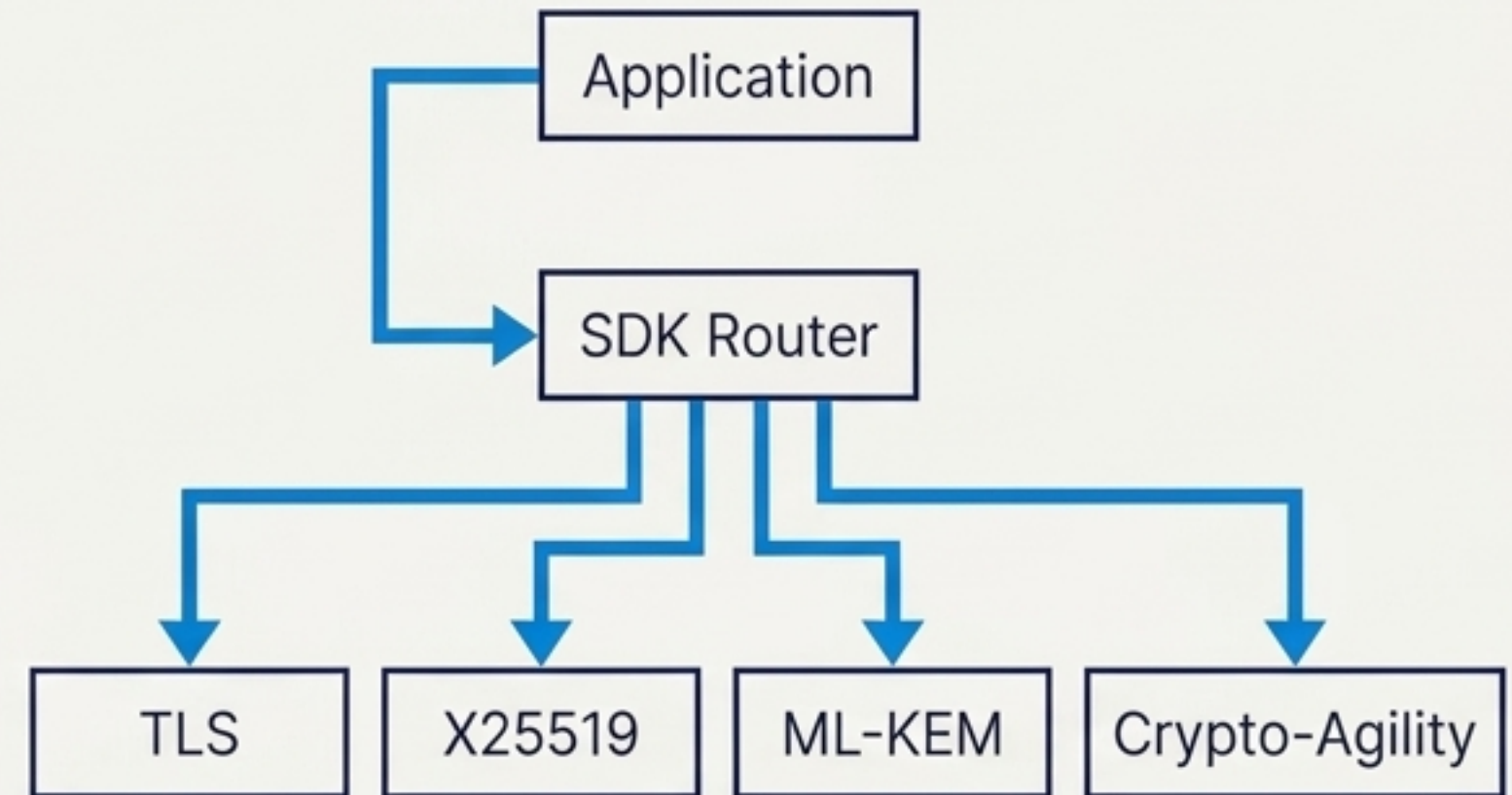
Requires extreme differentiation to beat open-source. Must be formally verified, constant-time, side-channel resistant, and hardware-optimized.



Revenue: Annual License + Support

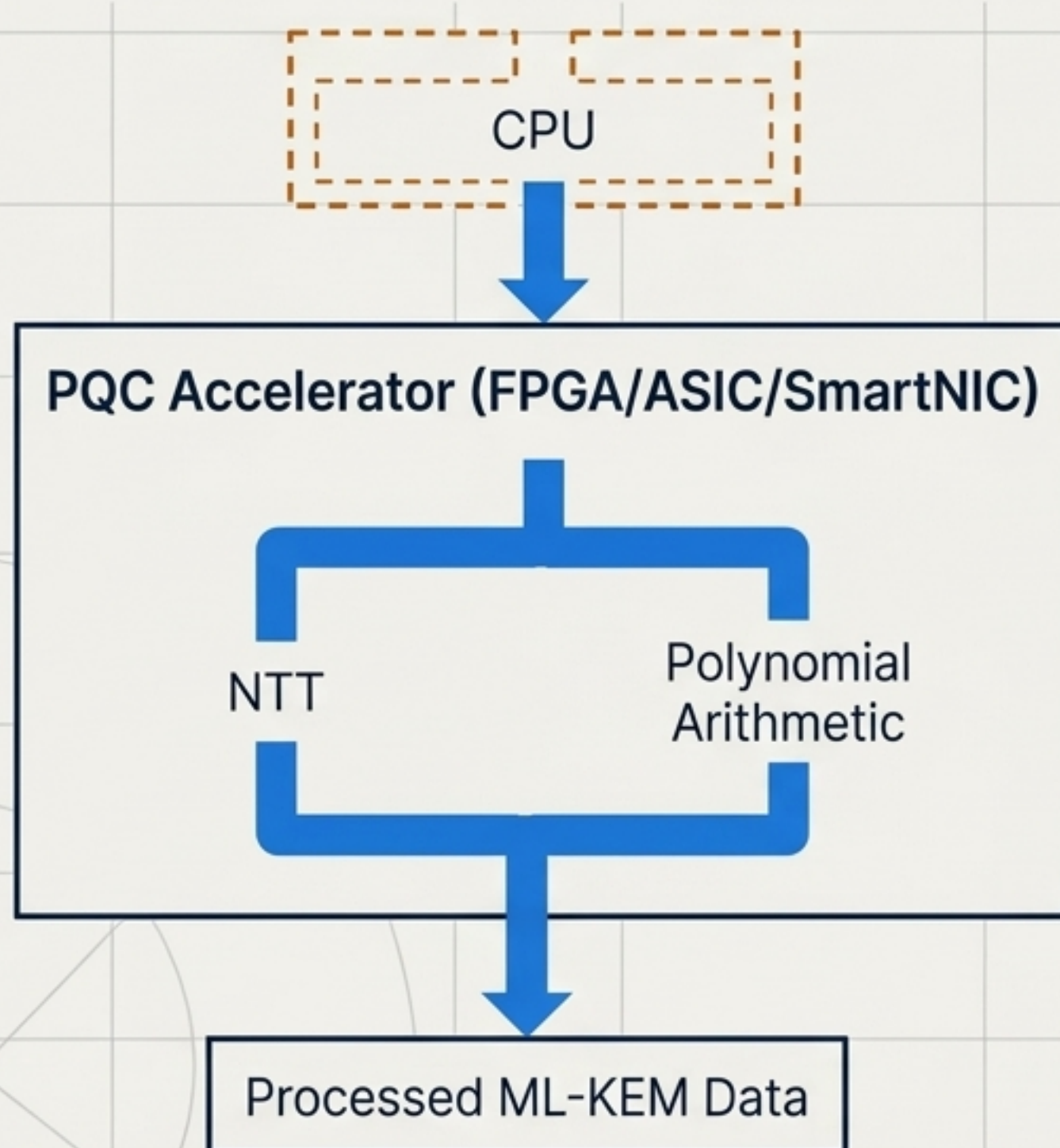
The SDK / API Model

The customer does not need to become a PQC expert. The API handles hybrid policies natively.



Revenue: API calls, per-device, or server licensing

Pathway 2: Heavy Infrastructure & Acceleration



PQC Hardware Security Modules (HSM/KMS)

Serious enterprise infrastructure for banks, defense, and cloud providers.

Merges ML-KEM/ML-DSA alongside legacy AES/RSA.

Revenue: Hardware + Software + Maintenance.

PQC Accelerators

Solves the computational expense of new algorithms.

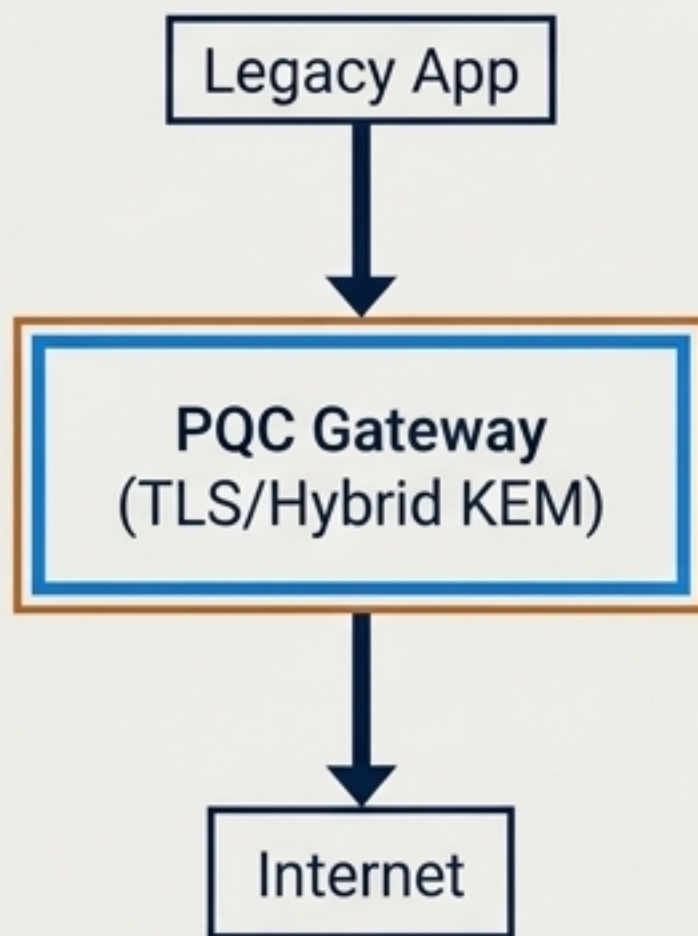
The Value Proposition

Customers buy higher throughput, lower latency, and lower CPU utilization—not just 'PQC'.

Pathway 3: The Seamless Integrators

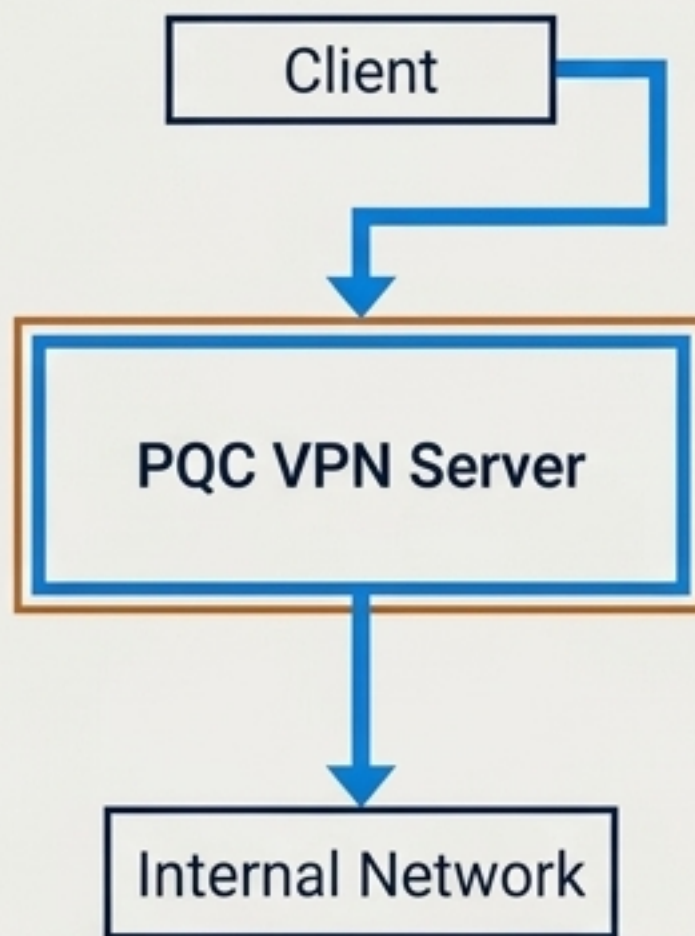
PQC Gateways

Zero-rewrite security. Legacy applications route through a gateway that handles hybrid KEM termination externally.



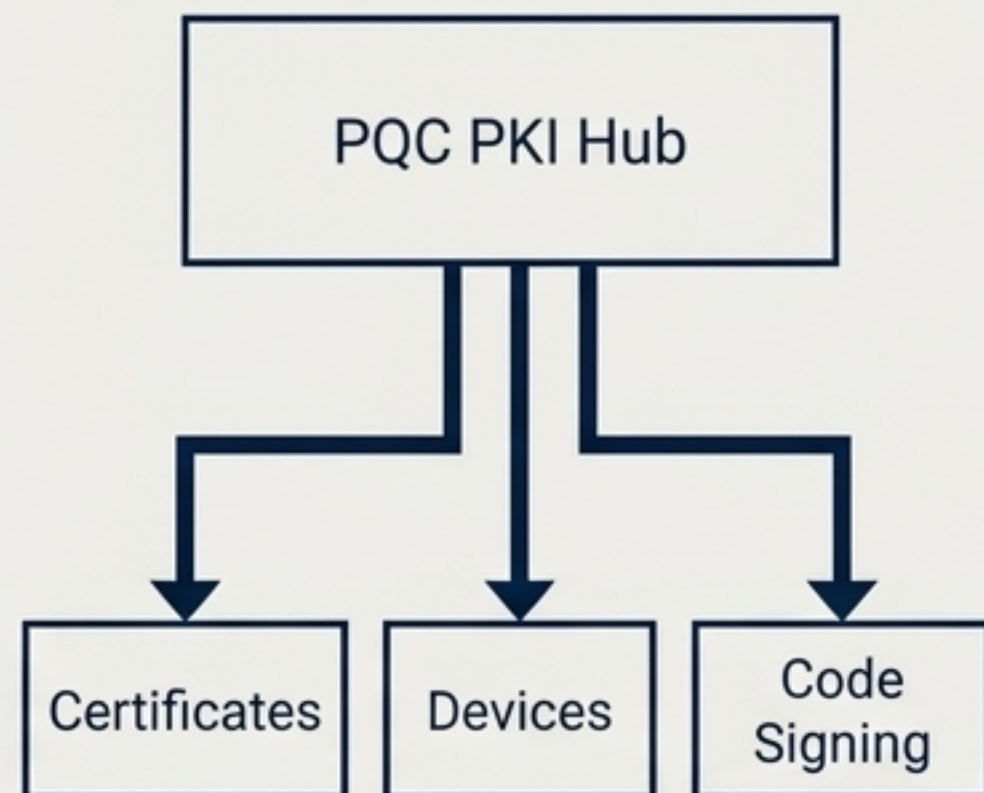
PQC VPNs

Protecting remote connections with quantum-resistant cryptography. Billed per user/device.

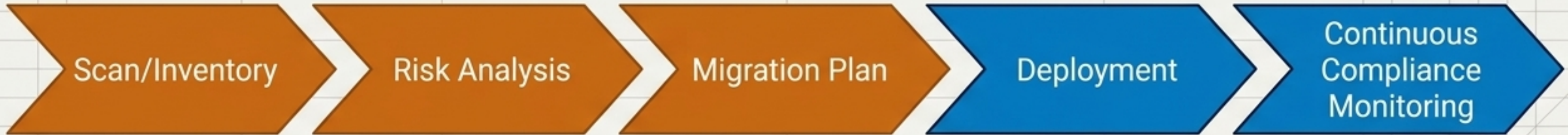


PQC PKI & Digital Identity

Managing ML-DSA/SLH-DSA certificate issuance, rotation, and revocation at scale.



Pathway 4: Strategic Entry Points (Discovery to Compliance)



CRYPTO INVENTORY

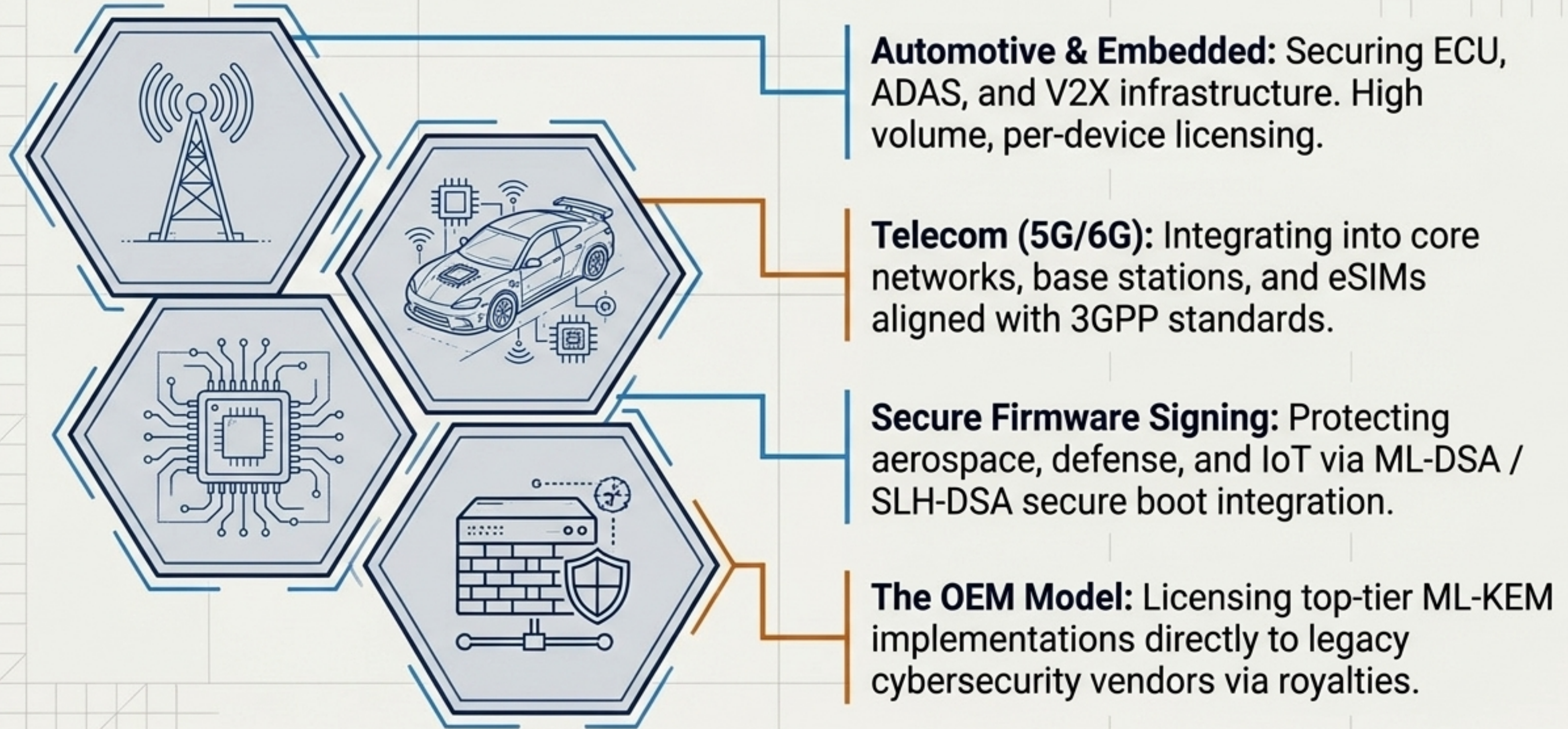
RSA-2048:	8,431
ECDSA P-256:	4,291
ML-KEM:	438
ML-DSA:	129

Consulting & Managed Services: Initial massive friction point. \$1M+ contracts to answer: "Where are we vulnerable?"

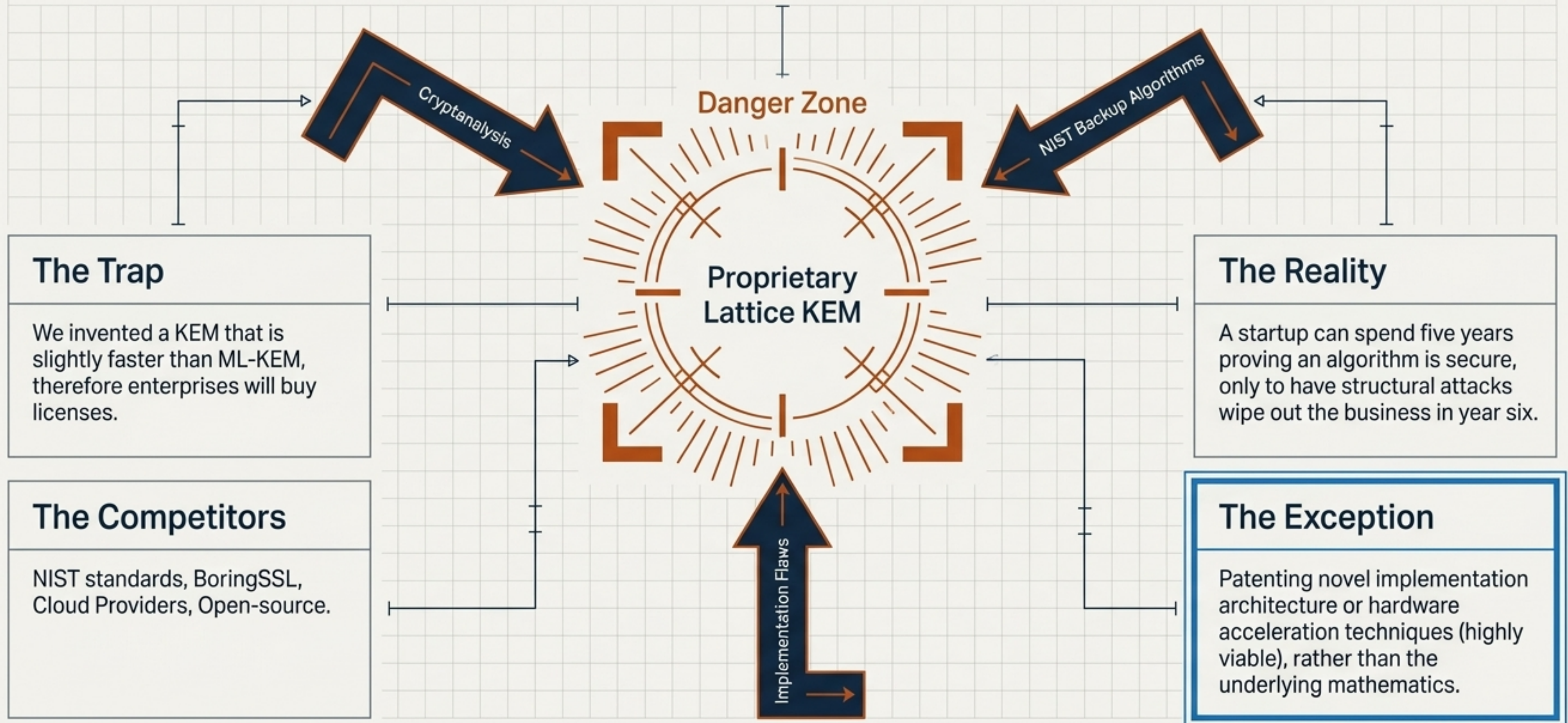
Migration SaaS Platforms: Productizing the consulting work. Scanning servers, cloud, and APIs to build a Crypto Bill of Materials (CBOM).

Compliance as a Product: Selling governance to CISOs. Demonstrating risk reduction and audit-ready migration status.

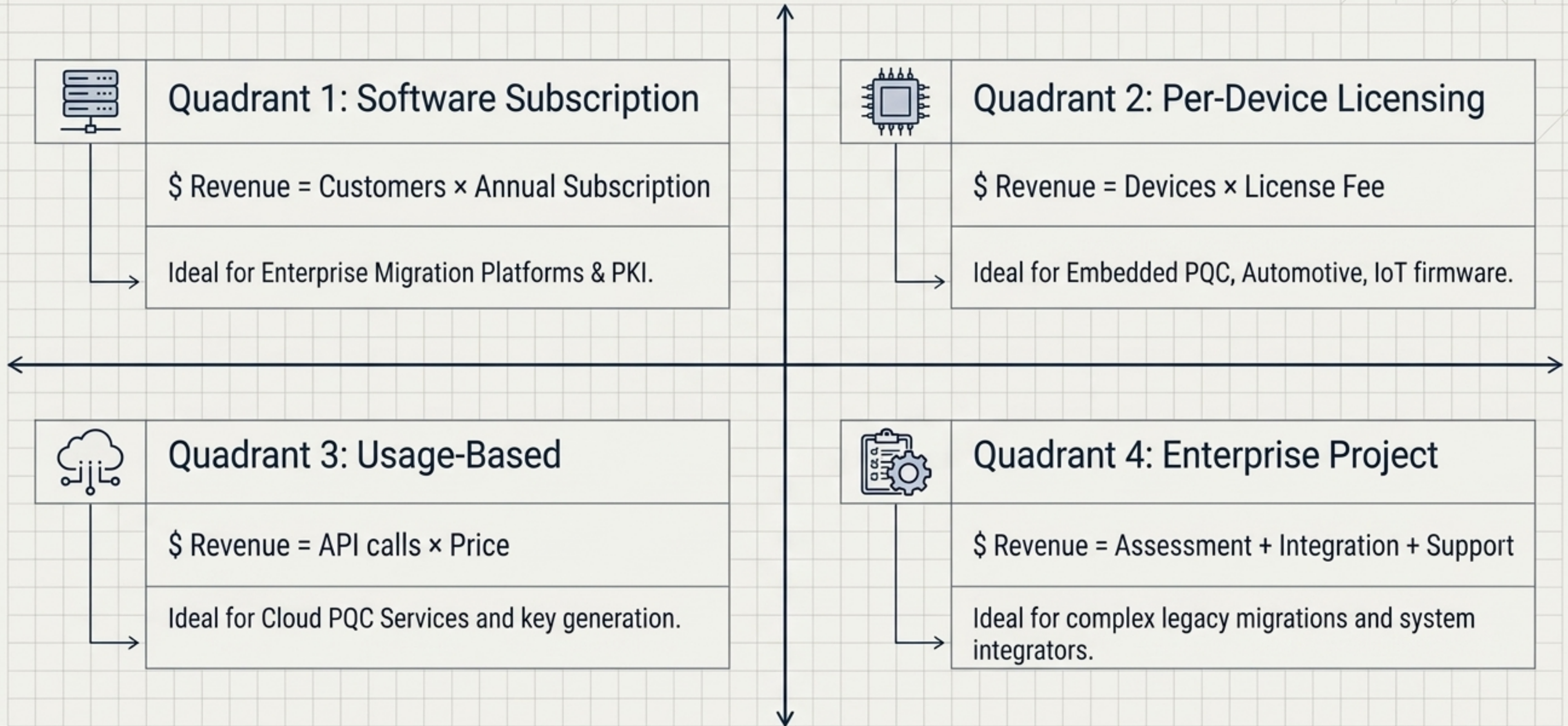
Pathway 5: Niche Verticals & OEM Licensing



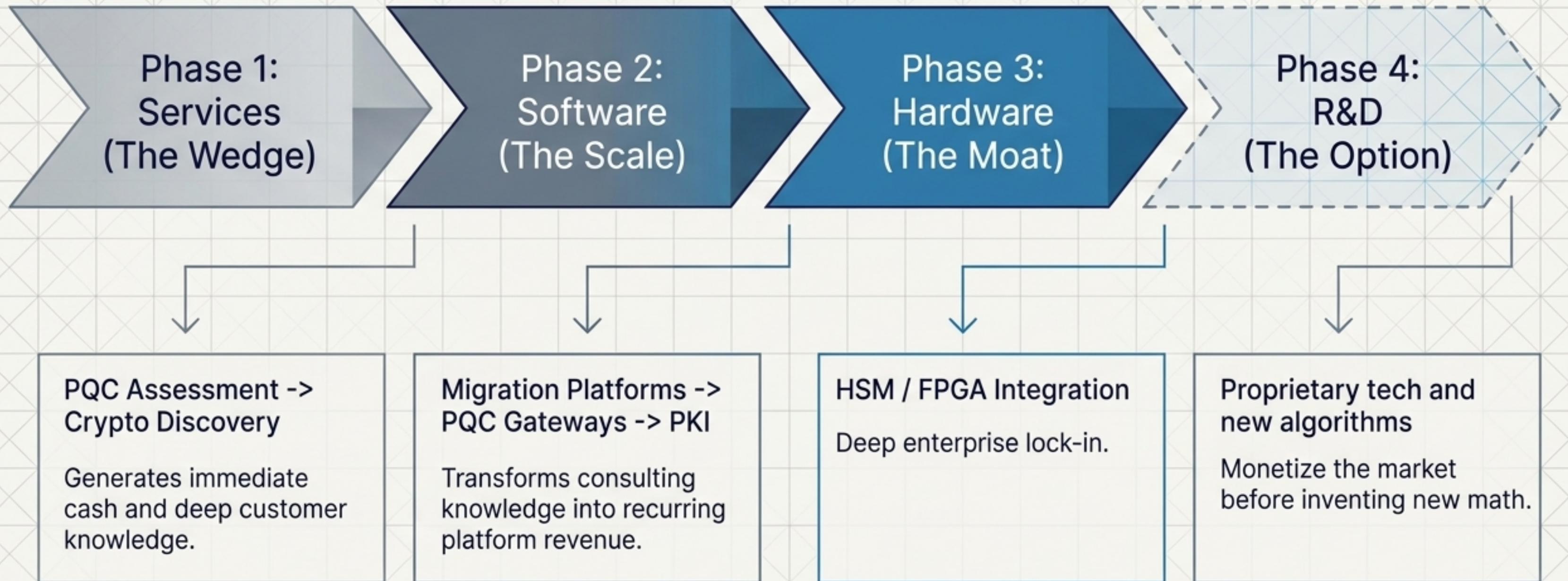
The Highest-Risk Bet: Selling Proprietary Algorithms



The Four Economic Engines of PQC

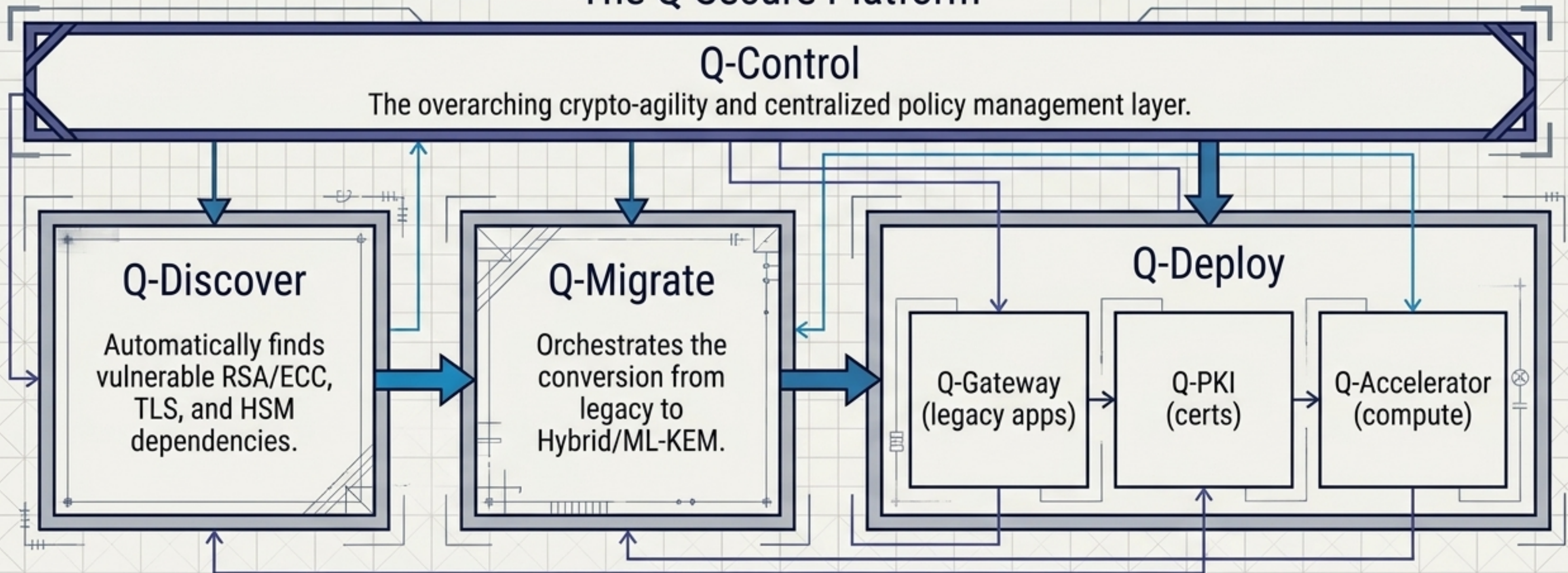


The Optimal GTM Sequencing



Blueprint for the Ultimate PQC Enterprise

The Q-Secure Platform



The Business Model: Platform Subscription + Deployment + Support + Hardware

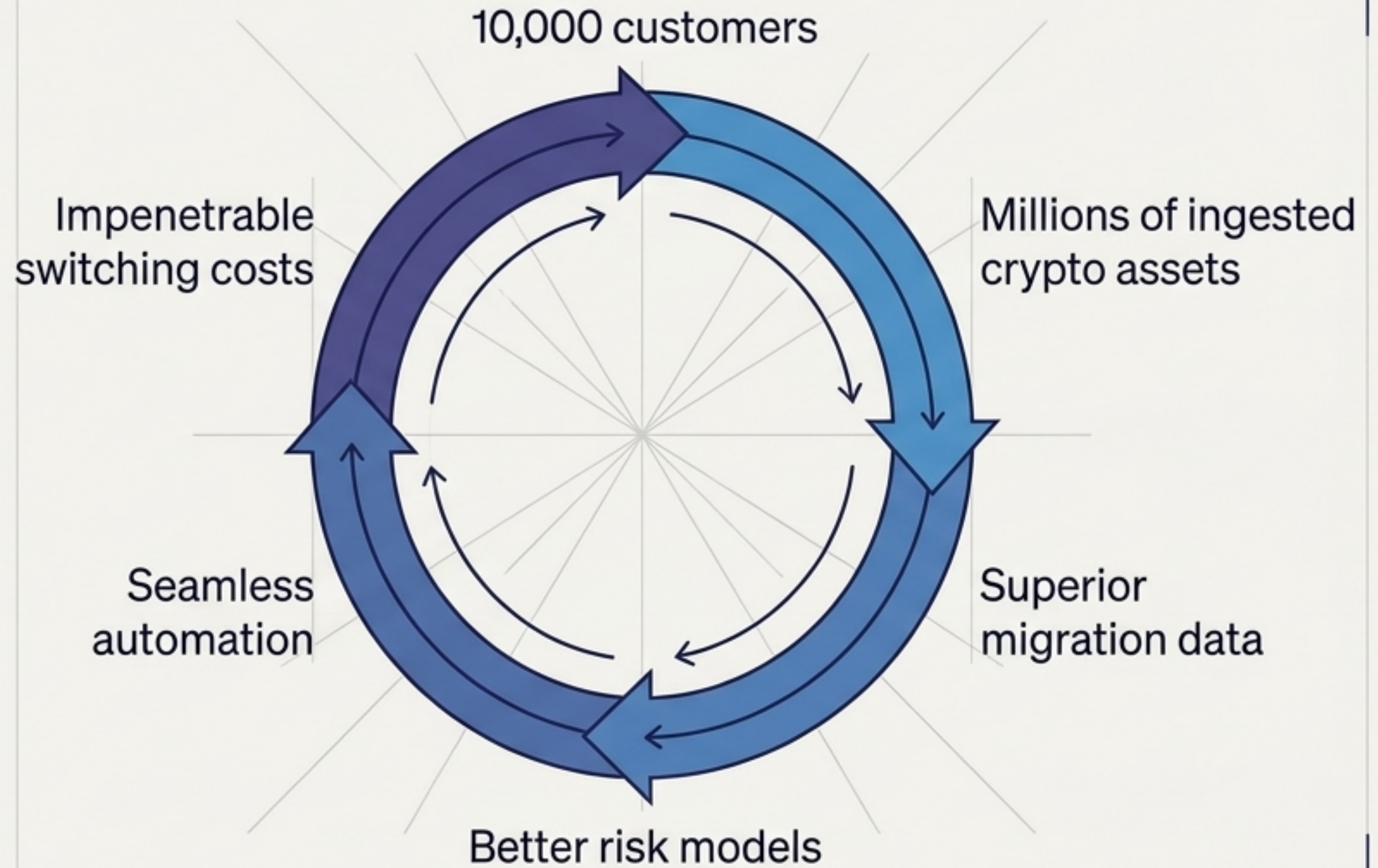
The PQC Moat is Built on Data, Not Secrecy

The Fallacy:

“Our moat is our knowledge of ML-KEM.”
(This is commoditized).

The Real Moat:

Crypto Discovery +
Migration Knowledge +
Enterprise Integrations +
Hardware Optimization.



The Evaluator's Rubric

Are you selling the algorithm itself?

YES

RED FLAG: Who has independently cryptanalyzed it?

NO

Are you selling an ML-KEM implementation?

YES

DIAGNOSTIC: What is the proprietary hardware/software wrap?

NO

Are you selling PQC solutions?

YES

INVEST/BUILD: Evaluate the Revenue Chain.

The Winning Formula: Customer -> Problem -> Product -> Deployment -> Recurring Revenue

Final Takeaway: A strong PQC startup is a cybersecurity business.
A weak one is a cryptography research bet wearing a sales badge.